

Cardiac Contractility Modulation (CCM) for Heart Failure (HF)

20.39 Links in PDF documents are not guaranteed to work. To follow a web link, please use the MCD Website.

Tracking Information

Publication Number

100-3

Manual Section Number

20.39

Manual Section Title

Cardiac Contractility Modulation (CCM) for Heart Failure (HF)

Version Number

1

Effective Date of this Version

10/28/2025

Implementation Date

04/06/2026

Description Information

Benefit Category

Outpatient Hospital Services Incident to a Physician's Service

Physicians' Services

Prosthetic Devices

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Cardiac Contractility Modulation (CCM) is used in the treatment of heart failure (HF).

Indications and Limitations of Coverage

B. Nationally Covered Indications

The Centers for Medicare & Medicaid Services (CMS) covers CCM for HF management under Coverage with Evidence Development (CED) when furnished according to a Food and Drug Administration (FDA) market-authorized indication and all of the following conditions are met:

1. Patient Criteria

Patients must meet the FDA market-authorized indications for use and remain symptomatic despite at least 3 months of optimized guideline-directed medical therapy (GDMT) as determined by the heart team prior to CCM implantation.

Patients are excluded from coverage if they meet any of the following criteria:

- *Meet any of the contraindications in the FDA labeling; or,*
- *Have had a heart transplant; or,*
- *Are younger than 18 years old*

2. CED Study Criteria

The CCM and related items and services are furnished in the context of a CMS-approved CED study. CMS-approved CED study protocols must: include only those patients who meet the criteria in section B.1; and include all of the following:

a) Primary outcomes of all-cause mortality, HF hospitalizations, or a composite of these, through a minimum of 24 months. Each component of a composite outcome must be individually reported.

b) An active comparator.

c) A care management plan that identifies members, roles, and responsibilities of the clinical team that performs the follow-up CCM patient management.

d) Design sufficient for subgroup analyses by:

- *Age (Stratify <65, 65-74, 75+)*
- *Other clinically important patient demographic factors*
- *Ischemic cardiomyopathy versus non-ischemic cardiomyopathy;*
- *No CRT, CRT (cardiac resynchronization therapy);*
- *Left ventricular ejection fraction (LVEF) $\geq 35\%$ versus $< 35\%$;*
- *Systolic blood pressure = median versus $<$ median;*
- *Diabetic vs non-diabetic*

e) In addition, CMS-approved CED studies must adhere to the following scientific standards (criteria 1-17 below) that have been identified by the Agency for Healthcare Research and Quality (AHRQ) as set forth in Section VI. of CMS' Coverage with Evidence Development Guidance Document, published August 7, 2024 (the "CED Guidance Document")

<https://www.cms.gov/medicare-coverage-database/view/medicare-coverage-document.aspx?mcid=38>

1. Sponsor/Investigator: The study is conducted by sponsors/investigators with the resources and skills to complete it successfully.

2. *Milestones: A written plan is in place that describes a detailed schedule for completion of key study milestones, including study initiation, enrollment progress, interim results reporting, and results reporting, to ensure timely completion of the CED process.*

3. *Study Protocol: The CED study is registered with ClinicalTrials.gov and a complete final protocol, including the statistical analysis plan, is delivered to CMS prior to study initiation. The published protocol includes sufficient detail to allow a judgment of whether the study is fit-for-purpose and whether reasonable efforts will be taken to minimize the risk of bias. Any changes to approved study protocols should be explained and publicly reported.*

4. *Study Context: The rationale for the study is supported by scientific evidence and study results are expected to fill the specified CMS-identified evidence deficiency and provide evidence sufficient to assess health outcomes.*

5. *Study Design: The study design is selected to safely and efficiently generate valid evidence of health outcomes. The sponsors/investigators minimize the impact of confounding and biases on inferences through rigorous design and appropriate statistical techniques. If a contemporaneous comparison group is not included, this choice should be justified, and the sponsors/investigators discuss in detail how the design contributes useful information on issues such as durability or adverse event frequency that are not clearly answered in comparative studies.*

6. *Study Population: The study population reflects the demographic and clinical diversity among the Medicare beneficiaries who are the intended population of the intervention, particularly when there is good clinical or scientific reason to expect that the results observed in premarket studies might not be observed in older adults or subpopulations identified by other clinical or demographic factors.*

7. *Subgroup Analyses: The study protocol explicitly discusses beneficiary subpopulations affected by the item or service under investigation, particularly traditionally underrepresented groups in clinical studies, how the inclusion and exclusion requirements effect enrollment of these populations, and a plan for the retention and reporting of said populations in the trial. In the protocol, the sponsors/investigators describe plans for analyzing demographic subpopulations as well as clinically relevant subgroups as identified in existing evidence. Description of plans for exploratory analyses, as relevant subgroups emerge, are also included.*

8. *Care Setting: When feasible and appropriate for answering the CED question, data for the study should come from beneficiaries in their expected sites of care.*

9. *Health Outcomes: The primary health outcome(s) for the study are those important to patients and their caregivers and that are clinically meaningful. A validated surrogate outcome that reliably predicts these outcomes may be appropriate for some questions. Generally, when study sponsors propose using surrogate endpoints to measure outcomes, they should cite validation studies published in peer-reviewed journals to provide a rationale for assuming these endpoints predict the health outcomes of interest. The cited validation studies should be longitudinal and demonstrate a statistical association between the surrogate endpoint and the health outcomes it is thought to predict.*

10. *Objective Success Criteria:* In consultation with CMS and AHRQ, sponsors/investigators establish an evidentiary threshold for the primary health outcome(s) to demonstrate clinically meaningful differences with sufficient precision.

11. *Data Quality:* The data are generated or selected with attention to provenance, bias, completeness, accuracy, sufficiency of duration of observation to demonstrate durability of health outcomes, and sufficiency of sample size as required by the question.

12. *Construct Validity:* Sponsors/investigators provide information about the validity of drawing warranted conclusions about the study population, primary exposure(s) (intervention, control), health outcome measures, and core covariates when using either primary data collected for the study about individuals or proxies of the variables of interest, or existing (secondary) data about individuals or proxies of the variables of interest.

13. *Sensitivity Analyses:* Sponsors/investigators will demonstrate robustness of results by conducting prespecified sensitivity testing using alternative variable or model specifications as appropriate.

14. *Reporting:* Final results are provided to CMS and submitted for publication or reported in a publicly accessible manner within 12 months of the study's primary completion date. Wherever possible, the study is submitted for peer review with the goal of publication using a reporting guideline appropriate for the study design and structured to enable replication. If peer-reviewed publication is not possible, results may also be published in an online publicly accessible registry dedicated to the dissemination of clinical trial information such as ClinicalTrials.gov, or in journals willing to publish in abbreviated format (e.g., for studies with incomplete results).

15. *Sharing:* The sponsors/investigators commit to making study data publicly available by sharing data, methods, analytic code, and analytical output with CMS or with a CMS-approved third party. The study should comply with all applicable laws regarding subject privacy, including 45 CFR § 164.514 within the regulations promulgated under the Health Insurance Portability and Accountability Act of 1996 (HIPAA) and 42 CFR, Part 2: Confidentiality of Substance Use Disorder Patient Records.

16. *Governance:* The protocol describes the information governance and data security provisions that have been established to satisfy Federal security regulations issued pursuant to HIPAA and codified at 45 CFR Parts 160 and 164 (Subparts A & C), United States Department of Health and Human Services (HHS) regulations at 42 CFR, Part 2: Confidentiality of Substance Use Disorder Patient and HHS regulations at 45 CFR Part 46, regarding informed consent for clinical study involving human subjects. In addition to the requirements under 42 CFR and 45 CFR, studies that are subject to FDA regulation must also comply with regulations at 21 CFR Parts 50 and 56 regarding the protection of human subjects and institutional review boards, respectively.

17. *Legal:* The study is not designed to exclusively test toxicity or disease pathophysiology in healthy individuals, although it is acceptable for a study to test a reduction in toxicity of a product relative to standard of care or an appropriate comparator. For studies that involve researching the safety and effectiveness of new drugs and biological products aimed at

treating life-threatening or severely debilitating diseases, refer to additional requirements set forth in 21 CFR § 312.81(a).

Consistent with section 1142 of the Social Security Act, AHRQ supports clinical research studies that CMS determines meet all the criteria and standards identified above.

C. Other Uses of CCM

- 1. CCM for HF management is not covered for patients outside of a CMS-approved study.*
- 2. Nothing in this NCD would preclude coverage of CCM for HF management through NCD 310.1 (Clinical Trial Policy) or through the Investigational Device Exemption (IDE) Policy.*

(This NCD last reviewed October 2025.)

Claims Processing Instructions

[TN 13538 \(Medicare Claims Processing\)](#) 

[TN 13672 \(Medicare Claims Processing\)](#) 

[TN 13716 \(Medicare Claims Processing\)](#) 

[TN 13806 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

13538

Coverage Transmittal Link

<https://www.cms.gov/files/document/r13538NCD.pdf> 

Revision History

05/2026 - Transmittal 13716 issued April 03, 2026, is being rescinded and replaced by Transmittal 13806, dated May 28, 2026, to revise the Pub.100-04 Claims Processing Manual (CPM) Table of Contents and chapter section/sub-section number(s), that were erroneously misnumbered. In the Pub 100-03 business requirement 1, the reference to the (CPM) is being updated to the correct chapter and section. There are no changes to the policy. All other information remains the same. ([TN 13806](#) ) (CR14311)

04/2026 - Transmittal 13672 issued March 09, 2026, is being rescinded and replaced by Transmittal 13716, dated April 3, 2026 to add Type of Bills (TOBs) 12X, 13X, 85X to the Claims Processing Manual (CPM) and a note regarding HCPCS codes C1824 and C1898 are not separately paid, as the payment is packaged and considered part of the eligible comprehensive procedure. Clarification language added on processing claims with HCPCS code K1030 by updating Pub. 100-04 Policy section, Business Requirements (BRs) 14311 - 04.3, adding BRs 14311 - 04.3.1, 14311 - 04.3.2 and 14311 - 04.4.1, this correction also updates Pub. 100-04 IOM. There are no changes to publication Pub 100-03. All other information remains the same. ([TN 13716](#) ) (CR14311)

03/2026 - Transmittal 13538 issued December 19, 2025, is being rescinded and replaced by Transmittal 13672, dated March 9, 2026, to add four (4) Place of Service (POS) codes to business requirements 14311-4.4 and 14311-4.11 and in the publication 100-04 manual attachment sections 416.2 and 416.4. This correction also updates the spreadsheet. There are no changes to publication 100-03. All other information remains the same. ([TN 13672](#)) (CR14311)

12/2025 - The purpose of this Change Request (CR) is to inform contractors that effective October 28, 2025, contractors shall pay claims for CCM. ([TN 13538](#)) (CR14311)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Cardiac Contractility Modulation \(CCM\) for Heart Failure \(CAG-00469N\)](#)

Additional Information

Other Versions			
Title	Version	Effective Between	
Cardiac Contractility Modulation (CCM) for Heart Failure (HF)	1	10/28/2025 - N/A	You are here